Course Code: **CSE1032**	**Course Title:** Linux Fundamentals	**TPC**	2	2	3
Version No.	x.xx				
Course Pre-requisites/ Co-requisites/ Anti-requisites	If any/ None				
Course Objectives:	1. To make the students familiar with Linux environment through common shell scripts 2.To make students understand the file structure in Linux and be able to manage files and directories				
Expected Outcome:	1.Understanding basic Linux commands and explore the Linux environment the utilities. 2.Develop shell scripts to perform complex tasks in Linux Environment 3.Understanding the file systems and manage file and directories using system calls 4.Implement various process management applications using C program				
Module 1	**Introduction**	**Assignment to be given/ Tools to be used**	**Module delivery (in hours 4)**		
Basic Concept of operating system, Features and Applications of LINUX, Difference between WINDOWs and LINUX, The LINUX Architecture, Functions of Kernel, Types of kernel, Types of Shells, Logging in with Username and password, The Command : date, tput clear, cal, who, ps, ls, cat, wc, echo, exit , LINUX and GNU					
Module 2	**The LINUX Architecture and Command usage**	**Assignment to be given/ Tools to be used**	**Module delivery (in hours) 3**		
The System Calls, Pattern Matching, Locating Commands, The Path, Internal and External Commands, Command Structure, Flexibility of Command Usage, man:Browsing the manual pages on-line, Understanding the man documentation, Further Help With man -k, different file and directory handling commands.					
Module 3	**The File System**	**Assignment to be given/ Tools to be used**	**Module delivery (in hours) 4**		
General Purpose Utilities: printf, mailx, passwd, uname, tty, stty, The File, The Parent-Child Relationship,pwd,cd ,Absolute and Relative Pathnames, ls:Listing Directory Contents, The LINUX File System, Handling Ordinary Files, Basic File Attributes.					
Module 4	**The vi Editor and The Shell**	**Assignment to be given/**	**Module delivery (in hours)**		

		Tools to be used	**5**
Vi Basics, Input mode, The ex Mode, Navigation, Editing Text, Undoing Last Editing Instructions, Repeating the last command. Redirection:The Three Standard Files, tee: Creating a Tee, Simple Filters: pr,head,tail,cut,paste,sort,uniq,tr,grep:searching for a pattern,Basic Regular Expression(BRE),Extended Regular Expressions(ERE) and egrep,sed:The Stream Editor.			
Module 5	**The Processes in Linux**	**Assignment to be given/ Tools to be used**	**Module delivery (in hours) 4**
Process Basics,ps:Process Status,System Processes(-e or -a),Process Creation, Process States and Zombies, nice:Jobs Execution with Low Priority, Killing Processes with Signals, Job Control, at and batch,cron, time. Customizing the Environment: Environment Variables, Aliases(bash and ksh),Writing Selected lines to a File(w),Text Editing.			
Module 6	**Module title** **Shell Programming**	**Assignment to be given/ Tools to be used**	**Module delivery (in hours) 10**
Introduction to shell and its types, variables, Operators, Performing arithmetic in shell scripts, Branching and Loop Control Structures, Arrays, Shell Substitutions, Shell Functions, some useful text manipulation sample programs.			
Module 7	**Module title** Contemporary Issues	**Assignment to be given/ Tools to be used**	**Module delivery (in hours)**
Contemporary developments in the area related to this course.			
Outline of the Assignment/ Project work/ Tools to be used/ Software used, etc. (if any):			
Textbook: **1.** 1. Sumitabha Das, UNIX Concepts and Applications, (4e), Tata McGraw Hill Publications, 2011.			
Reference book(s): 1. Neil Matthew, Richard Stones, "Beginning Linux Programming", 2016 5th Edition, Wiley Publications. 2. Robert Love, "Linux System Programming: Talking Directly to the kernel and C Library", 2015, Second Edition, O Reilly Media.			
Mode of Evaluation	Please refer to the trailing guidelines. Suitable part may be 'cut & pasted' here.		

Couse under OCNE?	Is CA under Open Class Notes Exam mode (OCNE)?: Yes/ No
Has PBL component	Yes/ No If Yes, indicate the weightage and whether it is individual or group
SPOC	Dr. Priyanka Chakraborty
Recommended by the Board of Studies on	BoS No. and held on xx-xx-xxxx
Date of Approval by the Academic Council	Xxth Academic Council held on xx-xx-xxxx

Course Code: CSE5065 (same as that of theory part)	**Course Title (same as theory part)** Linux Programming	**LPC**	**0**	**2**	**1**
Version No.	x.xx (same as that of theory part)				
Co-requisites (if any)					
Objectives:	1. To make the students familiar with Linux environment through common shell scripts 2.To make students understand the file structure in Linux and be able to manage files and directories 2.				
Expected Outcome:	1. Understanding basic Linux commands and explore the Linux environment the utilities. 2.Develop shell scripts to perform complex tasks in Linux Environment 3.Understanding the file systems and manage file and directories using system calls 4.Implement various process management applications using C program				
List of Indicative Experiments (minimum of 15 experiments/ activities/ meetings per credit):					
1. Installation of Software 2. Basic shell commands like tty, passwd, cp, mv, rm, mount, unmask, ps, kill. 3. Different File handling commands like mkdir, rmdir, ls, cmp, comm, chmod, chown. 4. Use the cat command to create a file and to display the file. 5. An interactive grep script that asks for a word and a file name and then tells how many lines contain that word.					

6. A shell script that takes a command-line argument and reports on whether it is directory, a file or something else.
7. A shell script that accepts one or more file name as arguments and converts all of them to upper case, provided they exists in the current working directory.
8. (a) A shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers. b) A shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
9. An interactive file-handling shell program (copying, Removing, renaming, or linking files).
10. A shell script which receives two file names as arguments for checking whether the two file contents are same or not.
11. A shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
12. An interactive script that ask for a word and a file name and then tells how many times that word occurred in the file.
13. A shell script to perform the following string operations: i.To extract a sub-string from a given string. ii.To find the length of a given string.
14. A shell script to print sum of individual digits of a number.
15. A shell script to find factorial of a given number.
16. A shell script to list all of the directory files in a directory.

Is Lab digital report available?	**Yes/ No**
Is individual lab session evaluation metrics available?	**Yes/ No**

Textbook (if required):

1. Neil Matthew, Richard Stones, "Beginning Linux Programming", 2016 5th Edition, Wiley Publications.

References (if required):

1. Sumitabha Das, UNIX Concepts and Applications, (4e), Tata McGraw Hill Publications, 2011.
2. Robert Love, "Linux System Programming: Talking Directly to the kernel and C Library", 2015, Second Edition, O Reilly Media.

Mode of Evaluation	Continuous. All individual sessions carry specific marks and the cumulative marks of all sessions will be for a total of 100 marks.
SPOC	Dr. Priyanka Chakraborty
Recommended by the Board of Studies on	BoS No. and held on xx-xx-xxxx
Date of Approval by the Academic Council	Xxth Academic Council held on xx-xx-xxxx